

Manolis PERROT

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Year of birth: 1997
Nationality: French

Expertise

Maths: Geometric Fluid Dynamics, Deterministic and Stochastic Modeling, Uncertainty Quantification, Scientific Computing.
Physics: Ocean and Atmosphere Turbulence, Subgrid Parameterizations for Climate models, Energy conservation, Convection, Physics-Dynamics coupling, Surface Waves.

Appointments

2025 – 2027	Postdoctoral researcher: funded by ANR AWARE , hosted by Bruno Deremble at Institut des Géosciences de l'Environnement , CNRS, Grenoble: <i>Impact of surfaces waves on ocean vertical mixing</i> .
2021 – 2025	PhD candidate researcher: supervised by Eric Blayo , Florian Lemarié at Laboratoire Jean-Kuntzmann , Université Grenoble-Alpes and Inria: <i>Modeling Oceanic and Atmospheric Convection: Energy, Uncertainties and Rotation</i> .
2020 – 2021 (10 months)	ENS pre-doctoral research assistant: internship supervised by Eric Blayo , Florian Lemarié and Etienne Mémin at Laboratoire Jean-Kuntzmann , Grenoble: <i>Stochastic modelling of air-sea turbulent fluxes</i> .
2020 (5 months)	M2 research assistant (Master's thesis): supervised by Klas Modin (Göteborg Univ., Sweden): <i>Some geometric aspects of ideal hydrodynamics</i> .
2018 (5 months)	Research assistant: M1 internship supervised by Antoine Venaille and Pierre Delplace at Physics Laboratory of ENS de Lyon: <i>Topological transition in stratified fluids</i> .
2017 (3 months)	Numerical PDE project supervised by Emmanuel Dormy : <i>A model of water waves amplification due to wind stress</i> .
2017 (4 months)	Bachelor's thesis supervised by Laurent Charles : <i>Egorov's theorem and mathematical quantization</i> .
2016 (6 months)	TIPE experiments and thesis for the ENS exam entrance: <i>Electrical resistivity tomography methods applied to the detection of mountain permafrost</i> .

Publications and preprints

Perrot M., Lemarié F., [Energetically Consistent Eddy-Diffusivity Mass-Flux Convective Schemes: 2. Implementation and Evaluation in an Oceanic Context](#), *Journal of Advances in Modeling Earth System* (2025).

Perrot M., Lemarié F., Dubos T., [Energetically Consistent Eddy-Diffusivity Mass-Flux Convective Schemes: 1. Theory and Models](#), *Journal of Advances in Modeling Earth System* (2025).

Modin K., Perrot, M.¹ [Eulerian and Lagrangian stability in Zeitlin's model of hydrodynamics](#), *Communications in Mathematical Physics* (2024) [[arXiv](#)].

Perrot M., Delplace P., Venaille A., [Topological transition in stratified fluids](#), *Nature Physics* (2019) [[arXiv](#)].

Education

2021 – 2025	ECOLE DOCTORALE MSTII, UNIVERSITÉ GRENOBLE ALPES, Grenoble, France. PhD in applied mathematics: <i>Modeling Oceanic and Atmospheric Convection: Energy, Uncertainties and Rotation</i> .
2016 – 2020	SORBONNE UNIVERSITÉ & ECOLE NORMALE SUPÉRIEURE (ENS) ² , Paris, France. Bachelor and Master of pure Mathematics , additional courses in physics and climate sciences.
2014 – 2016	CLASSES PRÉPARATOIRES STANISLAS (PCSI – PC*), Paris, France. Successful scientific preparation for the national entrance exams to the ENS, with major in mathematics, physics and chemistry.

Community service

2026	Reviewer for Environmental Fluid Mechanics (one review).
2025 – 2027	Member of Equality-Diversity-Inclusion commission of IGE.
2024 – 2025	Elected member of Academic Council and Research Commission of Université Grenoble-Alpes .
2021 – 2025	Member of Equality-Diversity-Inclusion commission of Lab. Jean Kuntzmann.

Awards

¹In the mathematical community, authors are listed in alphabetical order

²Selective french school preparing students to research or teaching career. Students are paid as civil servants during their 4-year tuition

2025

MSTII doctoral school 2025 Outstanding thesis.

Teaching & outreach

2021 & 2022	Teaching assistant in linear algebra for civil engineering students (Licence 2 – 60h).
2017	Organization of a workshop during a meeting for highschool girls in mathematics (Rendez-vous des Jeunes Mathématiciennes).
2016 – 2018 (2 years)	Oral examiner (<i>colleur</i>) in <i>Classe préparatoire</i> in Physics.

Events and Talks

May 2026	Speaker, GDR Theory and Climate , Paris, France: <i>Non-gradient parameterizations of oceanic convection: insights from APE theory</i> .
February 2026	Poster presentation, American Geophysical Union, Ocean Science Meeting , Glasgow, Scotland: <i>A Mass-Flux parameterization of Convection: Energy Conservation and Uncertainty Quantification</i> .
January 2026	Speaker, DRAKKAR Ocean Modelling workshop , Grenoble, France: <i>A Mass-Flux parameterization of Convection: Energy Conservation and Uncertainty Quantification</i> .
December 2025	Invited speaker, OSUG Turbulence Workshop , Grenoble, France: <i>Calibration of numerical models</i> .
October 2024	Speaker, Fine Scales in Numerical Ocean Models workshop, Brest, France: <i>Eddy-Diffusivity Mass-Flux convective parameterization: energy budgets, rotation effects and uncertainty quantification</i> .
May 2024	GDR DEPHY - Development and Evaluation of Physical Parameterizations workshop, Fréjus, France.
April 2024	Speaker, EGU Vienna , Austria: <i>Convective Mixing: an Energetically Consistent Non-Local Parameterization</i> .
March 2024	Invited speaker, PC6-7 TRACCS kickoff meeting Toulouse, France: <i>EDMF convective schemes: closed energy budgets and transport-map Bayesian estimation</i> .
January 2024	Speaker, Ateliers de Modélisation de l'Atmosphère Toulouse, France: <i>Energetically consistent coupling of EDMF and TKE schemes for Convection</i> .
December 2023	Invited speaker, MathsInFluid seminar ENS de Lyon, France: <i>Au-delà de la diffusion turbulente: la convection et ses paramétrisations</i> .
September 2023	Fluid Dynamics of Sustainable Environment summer school, Cambridge University, United-Kingdom.
July 2023	<i>Derivation and calibration of consistent Eddy-Diffusivity Mass-Flux parameterizations for ocean models</i> , talk at the PDEs on the Sphere conference, Grenoble, France.
June 2023	Attendee of Journée annuelle du GDR Théorie et Climat , Paris, France.
May 2023	GDR DEPHY - Development of Physical Parameterizations workshop, Fréjus, France.
April 2022	Short talk at the research school <i>Mathematical Advances in Geophysical Flows</i> , CIRM, Marseille, France: <i>Multi-fluid modelling for Ocean Deep Convection Parameterizations</i> .
November 2021	Attendee of the winter school <i>Numerical modelling of Ocean and Atmosphere</i> , Grenoble, France.
August 2021	Attendee of the <i>Workshop on Ocean-Atmosphere interaction</i> , Ecole de Physique des Houches, France.
June 2018	Attendee of the <i>Mathematical Physics Days : Quantum Chaos</i> , Lyon, France.

IT skills

TeX, Python, Bash, Fortran, Git, Unix.

Languages

French:	Mother tongue.
English:	Fluent.
Spanish:	Fluent.
Portuguese:	Good working knowledge.